BASIC ELECTRONICS

ISCED UNIT CODE: 0714 441 04A

TVET CDACC UNIT CODE: IT/CU/ICTA/CC/01/5/MA

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962

CONTINUOUS ASSESSMENT TEST (CAT 2)

TIME: 2 HOURS

TOTAL MARKS: 70

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INSTRUCTIONS TO THE CANDIDATE

i. *Answer all questions.*

ii. *Write your answers clearly and legibly.*

iii. *Draw well-labelled diagrams where required and show all working for calculations.*

1. A technician is examining electronic components for use in a control circuit.

a) Define semiconductor material. (2 marks)

b) Describe the structure of a silicon atom, indicating the number of valence electrons. (3 marks)

c) Distinguish between conductors and semiconductors based on the energy band. (4 marks)

2. A trainee is preparing semiconductor materials for the manufacture of electronic components.

a) Describe how doping produces P-type semiconductor material. (3 marks)

b) Describe how doping produces N-type semiconductor material. (3 marks)

c) State two differences between P-type and N-type semiconductor materials. (4 marks)

3. A technician connects a PN-junction diode to a DC power supply to determine its operating characteristics.

a) Draw a labelled diagram of a PN-junction diode under forward bias. (4 marks)

b) Explain how forward bias affects the depletion region and current flow in the diode. (4 marks)

c) Draw a labelled diagram of a PN-junction diode under reverse bias. (4 marks)

d) Explain how reverse bias affects the depletion region and current flow in the diode. (4 marks)

4. Using the theorems of Boolean algebra , prove the following identities: (6marks)

(i) $(A+\overline{B})\cdot C+(B+\overline{C})\cdot A+\overline{A}\cdot B=A+B+C$

(ii) $\overline{X}\cdot(X+Z)+\overline{Y}+Y\cdot Z=\overline{Y}+Z$

5. Simplify the following functions of Boolean algebra using the k-map. (6marks)

(i) $F(a,b,c,d) = \sum (0,1,2,8,9,10)$

(ii) $F(a,b,c,d) = \sum (0,1,4,6,9,13,14,15)$

6. A computer system uses different storage technologies to store data and instructions.

a) Classify each of the following as semiconductor, magnetic or optical memory:

i. Solid-state drive (SSD) (1 mark)

ii. Hard disk drive (HDD) (1 mark)

iii. DVD (1 mark)

b) Distinguish between internal storage and online storage. (2 marks)

c) State two differences between main memory and secondary storage. (2 marks)

7. An engineer is designing a digital control circuit that receives two inputs, A and B.

a) Draw the symbols for the AND, OR and NOT gates. (3 marks)

b) Draw the truth table for an XOR gate. (3 marks)

c) State one difference between NAND and NOR gates. (2 marks)

d) Simplify the following Boolean expression using Boolean algebra. (4 marks)

$$Y = (A \cdot B) + (A \cdot \overline{B})$$

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1a) Semi-conductor material is a material whose electrical conductivity lies between that of a conductor and an insulator.

b) A silicon atom has an atomic number of 14. It contains 14 protons and 14 electrons.

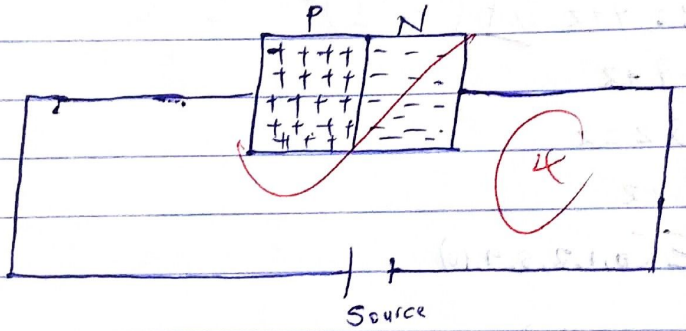
c) Valence and conduction Conductors have valence and conduction bands overlap while semiconductors have valence and conduction bands are separated by a small energy gap.

2a) Pure silicone or germanium is mixed with a trivalent impurity
The impurity has three valence electrons.
This creates holes in the material, making holes the majority carriers

b) Pure silicon or germanium is mixed with a pentavalent impurity.
The impurity has 5 valence electrons.
One extra electron becomes available for conduction, making electron the majority charge carriers.

c) P-type is doped with trivalent impurities while N-type is doped with pentavalent impurities.
P-type ~~is doped with~~ has holes as majority carriers while N-type has electrons are majority carriers.

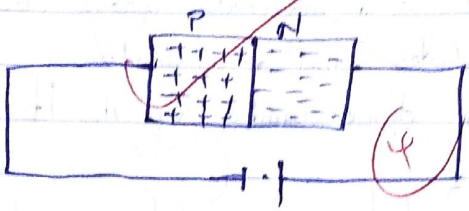
3. PN-junction diode (forwad bias)



b) The depletion region becomes narrower and the potential barrier is reduced.
The majority charge carriers cross the PN junction easily and a large current flows through the diode.

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c) PN junction diode (Reverse bias)



d) The depletion region becomes wider and the potential barrier increases. The majority charge carriers are prevented from crossing the junction. A very small leakage current flows through the diode until breakdown occurs.

4) $(A+\bar{B})\cdot C + (B+\bar{C})A + \bar{A}B = A+B+C$

$AC + \bar{B}C + AB + A\bar{C} + \bar{A}B$

$AC + A\bar{C} + AB + \bar{A}B + \bar{B}C$

$A(C+\bar{C}) + B(A+\bar{A}) + \bar{B}C$

$A+B+\bar{B}C$

$B+\bar{B}C = B+C$

$= A+B+C$

i) $\bar{x}(x+z) + \bar{y} + yz = \bar{y} + z$

$\bar{x}x + \bar{x}z + \bar{y} + \bar{y}z$

$\bar{x}z + \bar{y} + yz$

$\bar{x}z + (\bar{y} + yz)$

$\bar{y} + yz = \bar{y} + z$

$\bar{x}z + \bar{y} + z$

$z + \bar{x}z = z$

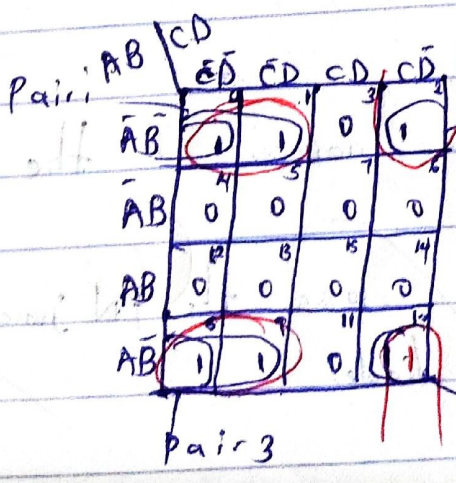
$\bar{y} + z$

$\bar{x}x = 0$

$\bar{x}z + \bar{x}z + \bar{y}z$

$\bar{x}x + (\bar{x} + \bar{y})z + \bar{y}$

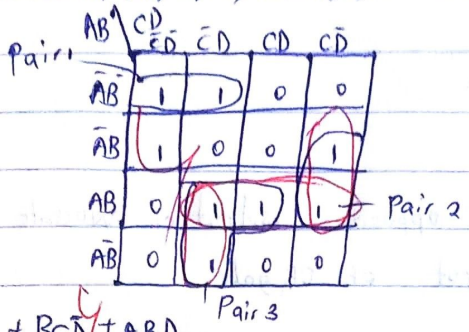
5) $F(a,b,c,d) = \Sigma(0,1,2,8,9,10)$



$\bar{A}\bar{B}\bar{C} + A\bar{B} + \bar{B}$

$F = \bar{b}\bar{c} + b\bar{d}$

ii) $F = (a, b, c, d) = \Sigma C(0, 1, 4, 6, 9, 13, 14, 15)$



$f = \bar{A}\bar{C}\bar{D} + \bar{B}\bar{C}D + B\bar{C}\bar{D} + ABD$

6) a) Solid State drive (SSD) - Semiconductor memory.

i) Hard disk drive (HDD) - magnetic memory.

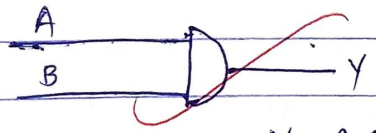
ii) DVD - Optical memory.

b) Internal storage stores data on a device's physical storage while online storage it stores data on remote servers through the internet.

c) The Main memory is faster to access while secondary storage is usually slower to access.

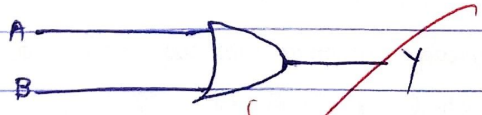
The main memory is volatile while the secondary storage is non-volatile.

7. AND GATE

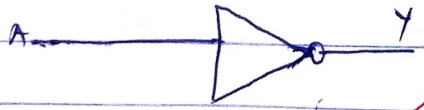


$Y = A \cdot B$

OR Gate



$Y = A + B$



$Y = \bar{A}$

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